GAS: A GLP App Store for Grassroots Social Contracts

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Abstract

GAS is a website that collects grassroots social contracts, shows how each contract is translated step by step into running code, and lets anyone use each contract live, as four people who each run the contract's platform. This document describes GAS and holds the example contracts written for it.

1 Introduction

A grassroots social contract is written in English, clause by clause. The GLP repository[1] carries a chain of tools, the cascade, that turns such a contract into its speech-act schemas, checks them, and compiles them into a platform that each person's device runs. GAS puts the contracts of that repository, and new ones, in one place where anyone can read them and try them.

2 The site

The site shows a graph of the contracts, the text of each contract, the layers of the cascade from the English text down to the platform's code, and a link to a live app for each compiled contract. It runs at `https://talmonlab.bgu.ac.il/glp`, and its code is at `https://github.com/nimrodtalmon/grassroots-contracts`.

3 The live server

One server process runs the compiled platform of each contract that someone opens. Each platform runs with four persons, Alice, Bob, Carol and Dave, in one GLP heap, and the web page sends each person's acts to that person's agent.

4 The example contracts

The contracts below were written for GAS. Each extends contracts of the GLP repository or of this collection, and cites their clauses without restating them. Each clause opens with its modality: ■ enforced, ♦ attested, ○ undertaken.

[1] `https://github.com/EShapiro2/GLP`

AGORA: Grassroots Barter

Alice, Bob and Carol are persons. AGORA extends GB. An instrument is a coin, a bond or an item.

1. ■ Making. Alice may make an Alice-item, one of a kind, which they then hold.
2. ■ Market. Alice may open a market under a name of their own not used before; Alice is its keeper and first member.
3. ■ Joining. Alice, the keeper of a market, may admit Bob, whom they know, by an act of both; Bob thereby agrees to come to know whoever posts to the market.
4. ■ Posting. Bob, a member of a market, may post to its wall an item they hold at the time, or something they want, each at most once, and so may Alice, its keeper; the keeper sees every post on their own wall and delivers it to every other member, the poster included, who thereby knows its poster.
5. ■ Withdrawal. Bob may withdraw a post of theirs; Alice, the keeper, removes it from their own wall and from the wall of every member it was delivered to, and Bob may not post it to that market again.
6. ■ Trading. GB's swap and escrow apply to items as to coins and bonds; a post is not withdrawn when its item is traded.

DAO: Grassroots DAO

Alice and Bob are persons. DAO extends GKK.

1. ■ Founding. Alice may found a DAO under a name of their own not used before; Alice is its steward and first member.
2. ■ Joining. Alice, the steward of a DAO, may admit Bob, whom they know and who is not yet a member, by an act of both.
3. ■ Proposing. A member of a DAO may make a proposal to it, each proposal once; its steward records it and delivers it to every member but the steward, the proposer included.
4. ■ Voting. A member who holds a proposal of their DAO may vote for or against it, once and before the steward decides it; the steward records the vote and delivers it to every member but the steward.
5. ■ Deciding. The steward may decide a proposal, passed or rejected, once; the steward delivers the decision to every member but the steward.
6. ■ Majority. The steward decides a proposal only once more than half of the DAO's members at that time have voted the same way on it, and passes it exactly when they voted for it.

LDAO: Liquid DAO

Alice and Bob are persons. LDAO extends DAO.

1. ■ Delegating. A member of a DAO, the steward included, may make another member, whom they know, their delegate in it, one at a time; the steward records the delegation.
2. ■ Revoking. A member may revoke their delegation at any time; the steward erases it.
3. ■ Counting. Once the delegate of a member who has not voted on a proposal has a vote on it counted, the steward counts the member as voting the same way, and the member cannot vote on it any more; so a vote passes down a chain of delegates, and a member whose chain ends with no vote, or runs into a cycle, is not counted.

FDAO: Funded DAO

Alice and Bob are persons. FDAO extends DAO and GB.

1. ■ Funding. A member of a DAO may give it coins they hold; its steward, who is not asked, keeps them apart as the DAO's, and they leave the DAO only as grants.
2. ■ Granting. The steward may pay the proposer of a passed proposal, once per proposal, any amount of the DAO's coins of one issuer; the proposer is not asked.
3. ○ Amount. The steward grants a passed proposal the amount it asks for, as far as the DAO's coins allow.

REC: Recommendations

Alice, Bob and Carol are persons. REC extends GKK.

1. ♦ Recommending. Alice may recommend Bob, whom they know, for something, a skill or a service, once for each thing; Bob is not asked and keeps the recommendation.
2. ■ Showing. Bob may show a recommendation they hold to Carol, whom they know, once; Carol then sees who recommended Bob and for what, but does not thereby come to know the recommender.
3. ■ Withdrawing. Alice may withdraw a recommendation; it is gone from Bob and from everyone Bob showed it to, and Alice may not make it again.

STAY: Grassroots Stays

Alice and Bob are persons. STAY extends GB.

1. ♦ Hosting. Alice, who can host guests at a place, may list it under a name not among their places.
2. ■ Booking. Bob, who knows Alice, and Alice may jointly decide that Bob stays at a place of Alice's on a night later than Alice's date, for a price Bob pays in Alice-coins; a place is booked at most once a night.
3. ■ Cancelling. Before the night, by Alice's date, Alice may cancel a stay, paying Bob back its whole price in Alice-coins; Bob is not asked.
4. ■ Reviewing. Once the night has come, by Bob's date, Bob may review their stay, once; Alice keeps the review and is not asked.

LIB: Grassroots Library

Alice and Bob are persons. LIB extends GB.

1. ♦ Shelving. Alice may put a thing of their own on their shelf, under a name they have not used before.
2. ■ Lending. Alice and Bob, who know each other, may jointly decide that Alice lends Bob a thing from Alice's shelf until a day later than Alice's date; it leaves the shelf, and Bob holds it but cannot lend it on.
3. ■ Returning. Bob may return a borrowed thing at any time; Alice is not asked, and it goes back on Alice's shelf.
4. ○ On time. Bob returns a borrowed thing by the agreed day.

JOINT: Joint Accounts

Alice and Bob are persons. JOINT extends GC.

1. ■ Opening. Alice and Bob, who know each other, may jointly open a joint account, under a name of Alice's not already used for one.
2. ■ Depositing. Either of them may move coins they hold into their joint account; the other is not asked.
3. ■ Withdrawing. Coins leave a joint account only by an act of both, to either of them.